THE BOUNDARY MAP OF THE PUNCTURED-LINE-BUNDLE SEQUENCE IS THE MAP ATTACHED TO THE FIRST DE RHAM CHERN CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The short exact sequence of the punctured line bundle and the residue map. Same chapter derham.tex, label lemma-the-complex-for-L-star (invoked at proof lines 1426, 1438, 1452 and 1491). It provides the shifted sequence $0 \to \Omega^\bullet_{X/S}[1] \to \Omega^\bullet_{L^\star/S,0}[1] \to \Omega^\bullet_{X/S} \to 0$ whose boundary is δ, and the local description of the residue map.
- The map attached to a de Rham cohomology class, and its uniqueness. Same chapter derham.tex, remark remark-cup-product-as-a-map (invoked at proof lines 1427 and 1459). It defines the map of the statement for a class ξ and states its uniqueness, which reduces the identification to checking the value on 1.
- Right differential graded modules over the de Rham complex and the derived category computation. Differential Graded Sheaves, Section \ref{sdga-section-misc} (invoked at proof line 1454). It shows that the boundary map, being a map of right differential graded $\Omega^\bullet_{X/S}$-modules, comes from a map in the derived category of such modules.
- Local trivializations and the Cech cocycle of an invertible sheaf. Same chapter derham.tex, Section \ref{section-first-chern-class} (invoked at proof line 1495). They present the class of $\mathcal{L}$ by the cocycle $\{f_{i_0i_1}\}$, which is what the final computation of the boundary uses.

Date: stacks-project:derham.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0FUF: sources/stacks-derham/pinned-original-source.tex, source lines 1348–1348. Navigation pointer for \ref{lemma-the-complex-for-L-star} used inside the counted span (line 1426); the argument that consumes it is retained in the span.

Source reference. Original remark 0FU7: sources/stacks-derham/pinned-original-source.tex, source lines 575–575. Navigation pointer for \ref{remark-cup-product-as-a-map} used inside the counted span (line 1427); the argument that consumes it is retained in the span.

Source reference. Original section 0FU3: sources/stacks-chow/tags.txt, source lines 19763–19763. Navigation pointer for \ref{sdga-section-misc} used inside the counted span (line 1454); the argument that consumes it is retained in the span.

Source reference. Original section 0FLE: sources/stacks-derham/pinned-original-source.tex, source lines 1035–1035. Navigation pointer for \ref{section-first-chern-class} used inside the counted span (line 1495); the argument that consumes it is retained in the span.

Lemma 0.1. *The "boundary" map $\delta : \Omega^\bullet_{X/S} \to \Omega^\bullet_{X/S}[2]$ in $D(X, f^{-1}\mathcal{O}_S)$ coming from the short exact sequence in Lemma 0FUF is the map of Remark 0FU7 for $\xi = c_1^{dR}(\mathcal{L})$.*

Proof. To be precise we consider the shift

$$0 \to \Omega^\bullet_{X/S}[1] \to \Omega^\bullet_{L^\star/S,0}[1] \to \Omega^\bullet_{X/S} \to 0$$

of the short exact sequence of Lemma 0FUF. As the degree zero part of a grading on $(L^\star \to X)_*\Omega^\bullet_{L^\star/S}$ we see that $\Omega^\bullet_{L^\star/S,0}$ is a differential graded $\mathcal{O}_X$-algebra and that the map $\Omega^\bullet_{X/S} \to \Omega^\bullet_{L^\star/S,0}$ is a homomorphism of differential graded $\mathcal{O}_X$-algebras. Hence we may view $\Omega^\bullet_{X/S}[1] \to \Omega^\bullet_{L^\star/S,0}[1]$ as a map of right differential graded $\Omega^\bullet_{X/S}$-modules on X. The map $\mathrm{Res} : \Omega^\bullet_{L^\star/S,0}[1] \to \Omega^\bullet_{X/S}$ is a map of right differential graded $\Omega^\bullet_{X/S}$-modules since it is locally defined by the rule $\mathrm{Res}(\omega' + \mathrm{d}\log(s) \wedge \omega) = \omega$, see proof of Lemma 0FUF. Thus by the discussion in Differential Graded Sheaves, Section 0FU3 we see that δ comes from a map $\delta' : \Omega^\bullet_{X/S} \to \Omega^\bullet_{X/S}[2]$ in the derived category $D(\Omega^\bullet_{X/S}, \mathrm{d})$ of right differential graded modules over the de Rham complex. The uniqueness averted in Remark 0FU7 shows it suffices to prove that $\delta(1) = c_1^{dR}(\mathcal{L})$.

We claim that there is a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{O}_X^* & \longrightarrow & E & \longrightarrow & \underline{\mathbf{Z}} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \mathrm{d}\log} & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \Omega^\bullet_{X/S}[1] & \longrightarrow & \Omega^\bullet_{L^\star/S,0}[1] & \longrightarrow & \Omega^\bullet_{X/S} & \longrightarrow & 0
\end{array}$$

where the top row is a short exact sequence of abelian sheaves whose boundary map sends 1 to the class of $\mathcal{L}$ in $H^1(X, \mathcal{O}_X^*)$. It suffices to prove the claim by the compatibility of boundary maps with maps between short exact sequences. We define E as the sheafification of the rule

$$U \longmapsto \{(s, n) \mid n \in \mathbf{Z},\ s \in \mathcal{L}^{\otimes n}(U) \text{ generator}\}$$

with group structure given by $(s, n) \cdot (t, m) = (s \otimes t, n + m)$. The middle vertical map sends (s, n) to $\mathrm{d}\log(s)$. This produces a map of short exact sequences because the map $Res : \Omega^1_{L^\star/S,0} \to \mathcal{O}_X$ constructed in the proof of Lemma 0FUF sends $\mathrm{d}\log(s)$ to 1 if s is a local generator of $\mathcal{L}$. To calculate the boundary of 1 in the top row, choose local trivializations s_i of $\mathcal{L}$ over opens U_i as in Section 0FLE.

On the overlaps $U_{i_0 i_1} = U_{i_0} \cap U_{i_1}$ we have an invertible function $f_{i_0 i_1}$ such that $f_{i_0 i_1} = s_{i_1}|_{U_{i_0 i_1}} s_{i_0}|^{-1}_{U_{i_0 i_1}}$ and the cohomology class of $\mathcal{L}$ is given by the Čech cocycle $\{f_{i_0 i_1}\}$. Then of course we have

$$(f_{i_0 i_1}, 0) = (s_{i_1}, 1)|_{U_{i_0 i_1}} \cdot (s_{i_0}, 1)|^{-1}_{U_{i_0 i_1}}$$

as sections of E which finishes the proof. □